

$$\begin{aligned}
 \text{Пр } \int_0^1 \frac{dx}{x^\alpha} &= \lim_{\omega \rightarrow 0+0} \int_\omega^1 \frac{dx}{x^\alpha} = \lim_{\varepsilon \rightarrow 0} \int_{0+\varepsilon}^1 \frac{dx}{x^\alpha} = \\
 &= \lim_{\varepsilon \rightarrow 0} \left(\begin{cases} \ln x, & \alpha = 1 \\ \frac{1}{x^{\alpha-1}(1-\alpha)}, & \alpha \neq 1 \end{cases} \right)_{0+\varepsilon}^1 = \lim_{\varepsilon \rightarrow 0} \left(\begin{cases} -\ln(0+\varepsilon), & \alpha = 1 \\ \frac{1}{1-\alpha} - \frac{1}{(0+\varepsilon)^{\alpha-1}(1-\alpha)}, & \alpha \neq 1 \end{cases} \right) \\
 &= \begin{cases} +\infty, & \alpha \geq 1 \\ \frac{1}{1-\alpha}, & \alpha < 1 \end{cases} \Rightarrow \text{нр. сс-се нрн } \alpha < 1 \\
 &\quad \text{р/сс нрн } \alpha > 1
 \end{aligned}$$

Отр Н.ч. по пер-му направлению нр.н.ч. I поре.
 Н.ч. от пер. ф-ции по втор-му ко-
 нсман функции нр.н.ч. II поре

$$\begin{aligned}
 \text{Пр } \int_{\frac{1}{2}}^{+\infty} f dx & \text{ - н.ч. } \pm p. \\
 \int_0^{\frac{1}{2}} f dx & \text{ - н.ч. } \pm p, \text{ где } x=0 \text{ - т.р. } \pm p
 \end{aligned}$$

$$(1) \quad] f \in R[a, b] \rightarrow \lim_{\omega \rightarrow b-0} \int_a^\omega f(x) dx = \int_a^b f(x) dx$$

$$\begin{aligned}
 \Delta \quad \int_a^\omega f(x) dx &= \Phi(\omega) \in C^0[a, b], \text{ т.к. } f \in R[a, b] \\
 \Rightarrow \lim_{\omega \rightarrow b-0} \Phi(\omega) &= \Phi(b) \Rightarrow \lim_{\omega \rightarrow b-0} \int_a^\omega f dx = \int_a^b f dx -
 \end{aligned}$$

- непрерывная функция. Δ

$$\text{Зам } \begin{cases} f \in R_{\text{loc}}[a, b) \text{ год} \\ f \text{ - оп на } [a, b) \end{cases} \Rightarrow f \in R[a, b]$$

Свойства н.ч.

⑦ (линейность)

$f, g \in R_{loc} [a, b)$

$$\text{Если } \begin{cases} \int_a^b f dx = cx \\ \int_a^b g dx = cx \end{cases} \Rightarrow \int_a^b (\alpha f + \beta g) dx = cx,$$

где $\alpha, \beta \in \mathbb{R}$, произвольные

$$\int_a^b (\alpha f + \beta g) dx = \alpha \int_a^b f dx + \beta \int_a^b g dx$$

$$\Delta \Rightarrow \int_a^b (\alpha f + \beta g) dx = \lim_{w \rightarrow b-0} \int_a^w (\alpha f + \beta g) dx =$$

$$= \lim_{w \rightarrow b-0} \left(\alpha \int_a^w f dx + \beta \int_a^w g dx \right) =$$

$$= \alpha \lim_{w \rightarrow b-0} \int_a^w f dx + \beta \lim_{w \rightarrow b-0} \int_a^w g dx =$$

$$= \alpha \int_a^b f dx + \beta \int_a^b g dx \in \mathbb{R} \Rightarrow$$

$$\Rightarrow \int_a^b (\alpha f + \beta g) dx = cx. \quad \Delta$$

Задача

$f, g \in R_{loc} [a, b)$,

$$\begin{cases} \int_a^b f dx = cx \\ \int_a^b g dx = px \end{cases}$$

$\Rightarrow$

$$\int_a^b (f + g) dx = p/cx$$

$$\Delta \text{ Отсюда: } \int_a^b (f + g) dx = cx - cx = 0$$

$$\Rightarrow \int_a^b \left(\underbrace{f+g}_{cs.} - \underbrace{f}_{cs.} \right) dx = cs.$$

$$\Rightarrow \int_a^b g dx = cs. \quad - \text{ n\textit{p}o\textit{y}b\textit{y}t\textit{e} - \textit{me} . \Rightarrow}$$

$$\int_a^b (f+g) dx = p / cs. \quad \Delta$$

Зам

$$\int_a^b f, g \in R_{loc} [a, b]$$

$$\int_a^b f dx = p / cs.$$

$$\int_a^b g dx = p / cs$$

$\Rightarrow$

$$\int_a^b (f+g) dx = \int_a^b cs - cs = p / cs.$$

Зп. $\int f = g = \frac{1}{x} \quad x \in [1, +\infty)$

$$\int_1^{+\infty} \frac{dx}{x} = p / cs.$$

$$\int_1^{+\infty} (f+g) dx = 2 \int_1^{+\infty} \frac{dx}{x} = p / cs.$$

$$\int_1^{+\infty} (f-g) dx = \int_1^{+\infty} 0 dx = 0 = cs.$$

$\textcircled{T}$

$$\int f \in R_{loc} [a, b] \Rightarrow \forall c \in (a, b)$$

$$\Rightarrow \int_a^b f dx = \int_a^c f dx + \int_c^b f dx,$$

$$\int_a^b f dx \sim \int_c^b f dx \quad cs. \quad \text{um} \quad p / cs. \quad \text{ognobp-m}$$

$$\Delta \int_a^w f dx = \int_a^c f dx + \int_c^w f dx, \quad a < c < w < b$$

$$\textcircled{E} \quad \int_a^b f dx + \int_a^b f dx \in \mathbb{R} \Rightarrow \int_a^b f dx - \text{const.} \in \mathbb{R}$$

$$\textcircled{T} \quad \left[\begin{array}{l} f, g \in R_{loc} [a, b) \\ \int_a^b f dx \in \overline{\mathbb{R}} \\ \int_a^b g dx \in \overline{\mathbb{R}} \\ f \leq g, x \in [a, b) \end{array} \right] \Rightarrow \int_a^b f dx \leq \int_a^b g dx$$

Δ T.K. $\int_a^u f dx \leq \int_a^u g dx \Rightarrow \lim_{u \rightarrow b-0} \int_a^u f dx \leq \lim_{u \rightarrow b-0} \int_a^u g dx \Rightarrow \int_a^b f dx \leq \int_a^b g dx \quad \Delta$

$\textcircled{T}$ φ -не непрерыв. по частям.

$$\left[\begin{array}{l} u, v \in \mathcal{D} [a, b) \\ u', v' \in R_{loc} [a, b) \end{array} \right] \Rightarrow$$

$$\int_a^b u v' dx = u v \Big|_a^b - \int_a^b v u' dx,$$

где $u v \Big|_a^b = \lim_{u \rightarrow b-0} u(u) v(u) - u(a) v(a),$

$$\text{или} \quad \int_a^b u dv = u v \Big|_a^b - \int_a^b v du, \quad \text{причем} \quad \textcircled{*}$$

$\textcircled{*}$ верно, если $\exists b \in \mathbb{R}$ такое что g не имеет на $[a, b]$ предела

$$\Delta \quad \text{T. K.} \quad \int_a^w u \, dv = uv \Big|_a^w - \int_a^w v \, du \quad \Rightarrow$$

$$\lim_{w \rightarrow b-0} \int_a^w u \, dv = \lim_{w \rightarrow b-0} (u(w)v(w) - u(a)v(a) - \lim_{w \rightarrow b-0} \int_a^w v \, du) \Rightarrow \int_a^b u \, dv = uv \Big|_a^b - \int_a^b v \, du \quad \Delta$$

$\textcircled{T} \quad \exists f \in C^0[A, B],$
 $x = \varphi(t): [\alpha, \beta] \rightarrow [A, B],$
 $\varphi \in \mathcal{D}[\alpha, \beta]$
 $\varphi' \in R_{loc}[\alpha, \beta]$
 $\exists \lim_{t \rightarrow \beta-0} \varphi(t) \in \overline{\mathbb{R}} \quad \Rightarrow$

$$\int_{\varphi(\alpha)}^{\varphi(\beta-0)} f(x) \, dx = \int_{\alpha}^{\beta} f(\varphi(t)) \varphi'(t) \, dt,$$

moreover, even $\exists$ open $b \in \overline{\mathbb{R}}$, to $\exists$ graphs
 of $\overline{\mathbb{R}}$.